

Univariate Calculus & Taylor Series

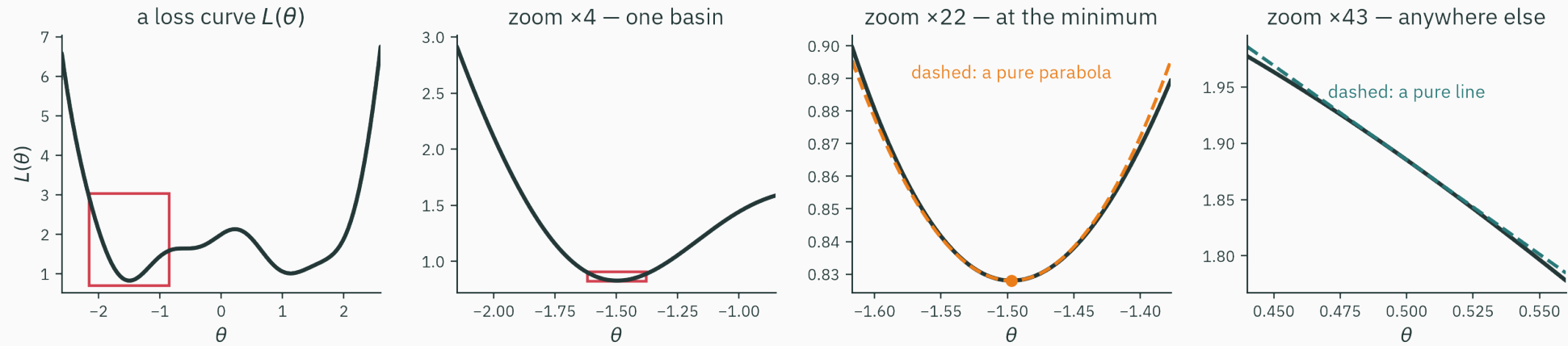
Every smooth function is locally a line

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Mathematical Foundations for AI · IIT Gandhinagar

Local linear and quadratic models of a loss

Zoom in on any loss curve



Wiggly from far away. Near a nondegenerate minimum ($\mathcal{L}'' > 0$), a **parabola** appears. At a point with nonzero slope, a **line** appears. Flat higher-order points such as x^4 are an important exception.

Guess before we continue: why a *line* almost everywhere, but a *parabola* exactly at the bottom? The answer is one formula — you'll derive it today.

Today's one idea

The derivative is the slope in a function's best local affine approximation. Taylor series systematically upgrades that local line to higher-degree polynomial approximations.

Why ML cares: training never sees the whole loss — only a **tiny neighbourhood** of the current weights. Local replacements are the **only** map it has.

Three questions, in order:

1. What is the best local **line**? → the derivative
2. Can we do better than a line? → Taylor's parabola, cubic, ...
3. Where does the local approximation **break**? → convergence radius

Learning outcomes

By the end of this lecture you will be able to:

1. Read $f'(x_0)$ as a **sensitivity**: Δ_{output} per Δ_{input} — and estimate it from a numeric table.
2. Use the **local line** $f(x_0 + h) \approx f(x_0) + f'(x_0) h$ to predict function values.
3. Apply the **chain rule** as ratio bookkeeping through a composition.
4. Classify flat points with the **second derivative** — curvature and its sign.
5. Build the **Taylor polynomial** of a function and bound its error (★ $\cos x$ at 0).
6. Explain the **convergence radius** — why a Taylor series can fail far from its base point.

A derivative is a sensitivity

Nudge the input, watch the output

Take $f(x) = x^2$ at $x_0 = 3$, so $f(3) = 9$. Nudge x and **measure the response**:

nudge h	$f(3 + h)$	response Δf	ratio $\Delta f/h$
1	16	7	7
0.1	9.61	0.61	6.1
0.01	9.0601	0.0601	6.01
0.001	9.006001	0.006001	6.001

The response shrinks with the nudge — but the **ratio** stabilizes: 7, 6.1, 6.01, 6.001, ...

The ratio is heading somewhere: 6. That destination is the derivative $f'(3)$.

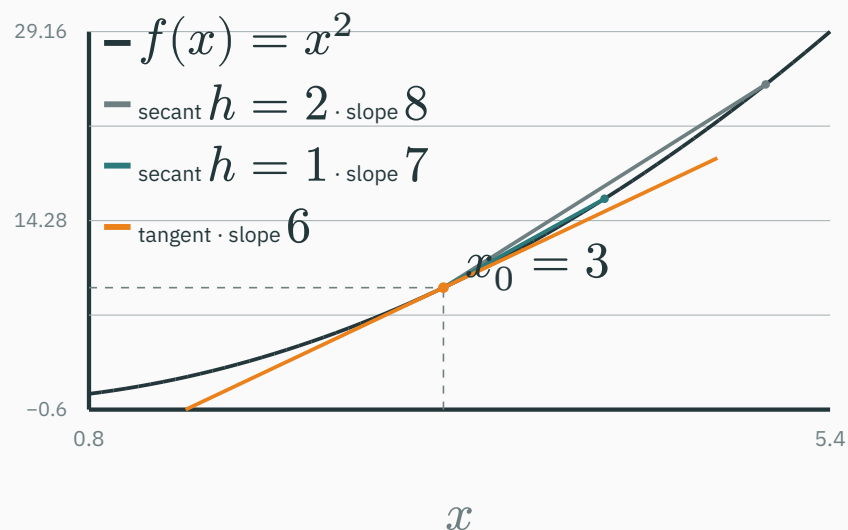
The nudge table, in code

```
f = lambda x: x**2
for h in [1, 0.1, 0.01, 0.001]:
    print(h, (f(3 + h) - f(3)) / h)
# 1          7.0
# 0.1        6.1000000000000012
# 0.01       6.0099999999999849
# 0.001      6.0009999999999479
```

L2 callback — see the garbage digits (...012, ...849)? That's floating-point rounding leaking into $\Delta f/h$. Shrink h too far and the noise **wins**.

The full tradeoff — truncation error vs rounding error in numerical derivatives — is worked out in L10. Today we take the **exact** route instead: limits.

The picture: secants tilting into the tangent



Each row of the table is a **secant line** through $(3, 9)$ and $(3 + h, f(3 + h))$. As $h \rightarrow 0$ the secants tilt into a limiting line: the **tangent**.

Only now, the symbols

The number the ratios approach gets a name:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

- The fraction is the **table**: response over nudge — an average rate across $[x_0, x_0 + h]$.
- The limit is the **stabilized ratio**: the instantaneous rate *at* x_0 .

Notation zoo, all the same object: $f'(x_0)$, $\frac{df}{dx}(x_0)$, $\dot{x}$ (physics). From L8 on, ∂ joins the zoo.

Read derivatives as sensitivities

$f'(x_0)$ answers one question: **if the input moves a little, how hard does the output move?**

function	f'	meaning
position $\rightarrow$ time	≈ 20 m/s	speed
revenue $\rightarrow$ price	-300 ₹/₹	raise price ₹1, lose ₹300
loss $\rightarrow$ one weight	-0.8	raise w by ε , loss drops 0.8ε

The last row is the local sensitivity that a standard first-order training method uses for this weight — millions of times per run (L17).

$f'(x_0) \approx \frac{\Delta \text{ output}}{\Delta \text{ input}}$ — a plain ratio of small changes. Everything else in this course is bookkeeping around that ratio.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/derivative-tangent

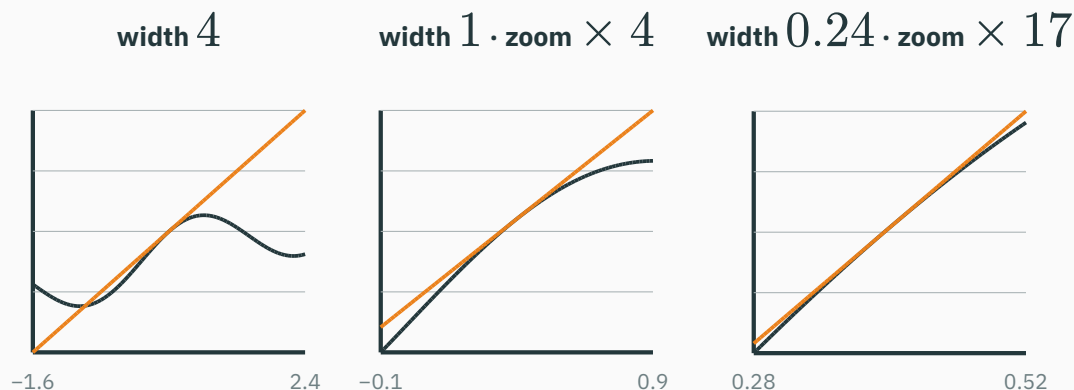
Drag the base point along x^2 , $\sin x$, or $x^3 - x$; shrink the secant step h and watch the average slope converge onto the derivative — then zoom until the curve straightens into its tangent line.

The two sliders are exactly today's two sections: **secant** → **tangent** (the limit) and **zoom** → **straight** (the local line).

A derivative is a local line

Zoom in: every smooth curve straightens

$f(x) = \sin 2x + x/2$ near $x_0 = 0.4$ — same curve, same dashed tangent, three window widths:



At $\text{zoom} \times 17$ the curve and its tangent are **indistinguishable**. “Smooth” *means* this: zoom far enough and there is only one line left.

The local replacement formula

Rearrange the derivative's definition: for small h , the ratio $\approx f'(x_0)$, so

$$f(x_0 + h) \approx \underbrace{f(x_0)}_{\text{where you are}} + \underbrace{f'(x_0)}_{\text{sensitivity}} \cdot h$$

This is a **trade**: swap a function you can't handle for the line you can.

Near x_0 , replace f by the line through $(x_0, f(x_0))$ with slope $f'(x_0)$. That replacement is what “derivative” is *for*.

Estimate $\sqrt{4.1}$ with the local line

$f(x) = \sqrt{x}$, base point $x_0 = 4$ (where $\sqrt{x}$ is exact): $f(4) = 2$, $f'(x) = 1/(2\sqrt{x})$, so $f'(4) = 0.25$.

$$\sqrt{4.1} \approx 2 + 0.25 \times 0.1 = 2.025$$

Truth: $\sqrt{4.1} = 2.02485\dots$ — the line is off by 0.00015, about **0.008%**, for zero effort.

This is how calculators, game engines, and hand computers before them evaluate functions: keep a few exact anchor points, **replace by lines (or short polynomials) in between.**

What “best line” means

Many lines pass through $(x_0, f(x_0))$. The tangent is the one whose error **dies faster than h** . For $f(x) = x^2$ at 3, the tangent’s error is exactly h^2 :

h	error = $f(3 + h) - (9 + 6h)$	error / h
0.1	0.01	0.1
0.01	0.0001	0.01
0.001	0.000001	0.001

Shrink h by $10 \times \rightarrow$ error shrinks $100 \times$; the **relative** error vanishes.

Pick any other slope $m \neq 6$: the error stays $\approx (6 - m)h$ — proportional to h , never negligible. Only the tangent’s error is superlinear. That is the sense in which it is “best”.

A sensor function satisfies $f(2) = 5$ and $f'(2) = -3$. Best quick estimate of $f(2.1)$?

A. 4.7

B. 5.3

C. 4.97

D. 2.0

Correct: A — $f(2.1) \approx 4.7$

Why: Local line: $f(2 + 0.1) \approx f(2) + f'(2) \times 0.1 = 5 + (-3)(0.1) = 4.7$. B flips the sign of the derivative; C uses $h = 0.01$; D subtracts f' without multiplying by the displacement h .

Rules, and the chain that runs ML

The rules you already know

School calculus, on one slide — this course **refreshes**, it does not reteach:

$(x^n)' = nx^{n-1}$	$(e^x)' = e^x$	$(\ln x)' = 1/x$
$(\sin x)' = \cos x$	$(\cos x)' = -\sin x$	$(af + bg)' = af' + bg'$
$(fg)' = f'g + fg'$	$(f/g)' = \frac{f'g - fg'}{g^2}$	$(f(g(x)))' = f'(g(x))g'(x)$

Only the last cell carries today's weight. Most ML losses are evaluated through a composition of operations, so the chain rule gets its own treatment.

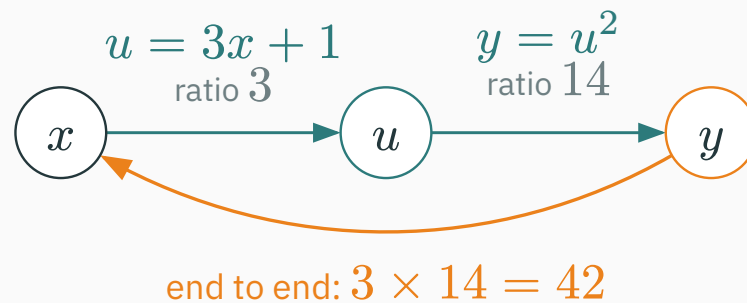
The chain rule is ratio bookkeeping

$y = (3x + 1)^2$ — a two-stage pipeline $x \rightarrow u = 3x + 1 \rightarrow y = u^2$. Nudge $x = 2$ by $\Delta x = 0.01$ and **track the nudge through the pipeline**:

stage	value	nudge arriving	local ratio
x	$2 \rightarrow 2.01$	$\Delta x = 0.01$	—
$u = 3x + 1$	$7 \rightarrow 7.03$	$\Delta u = 0.03$	$\Delta u / \Delta x = 3$
$y = u^2$	$49 \rightarrow 49.4209$	$\Delta y = 0.4209$	$\Delta y / \Delta u = 14.03$

End-to-end sensitivity: $\Delta y / \Delta x = 42.09 \approx 3 \times 14.03$. Each stage **multiplies the nudge by its local ratio** — the bookkeeping is just: multiply the ratios.

The chain rule, in symbols

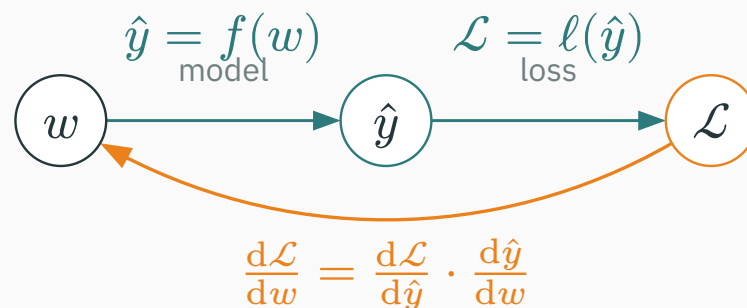


$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \text{here: } 2u \cdot 3 = 2(7) \cdot 3 = 42$$

In Leibniz notation the du 's **visually cancel** — that's the ratio bookkeeping made typographical.

Evaluate the outer ratio **at the inner value**: $dy/du = 2u$ is read at $u = 7$, not at $x = 2$. Forgetting this is the classic chain-rule bug.

Why ML is one long chain



A real network is this picture **a hundred stages deep**: weights $\rightarrow$ layer $\rightarrow$ layer $\rightarrow$... $\rightarrow$ loss. Training needs $d\mathcal{L}/dw$ for **every** weight.

**Same rule, industrial scale: multiply local ratios along the pipeline.
Doing that bookkeeping *efficiently* for millions of weights is
backpropagation — L10–L11.**

Flat points and curvature

Where the local line goes flat

If the best local line has **slope zero**, nudging cannot help you (to first order): x_0 is a **critical point**, candidate max or min. Worked: $f(x) = x^3 - 3x$:

$$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = -1 \text{ or } x = 1$$

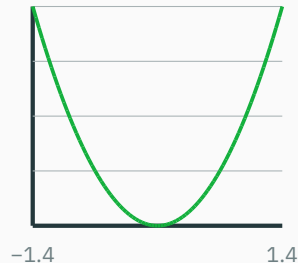


Both flat points have $f' = 0$ — the first derivative **cannot tell them apart**. We need the next order of information.

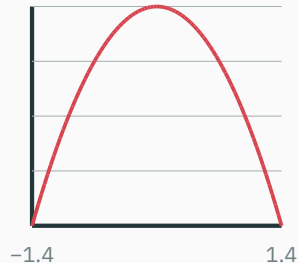
The second derivative is curvature

f'' is the derivative *of the slope*: is the local line **tilting up or down** as you pass through?

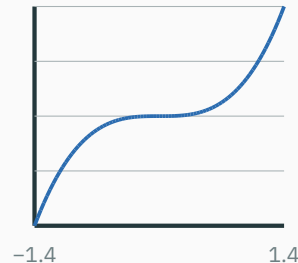
$$f'' > 0 \cdot \text{smile}$$



$$f'' < 0 \cdot \text{frown}$$



$$f'' = 0 \cdot \text{undecided}$$



Sign gallery: **smile** holds water (slope increasing), **frown** spills it (slope decreasing), zero curvature at a point settles nothing by itself.

The second-derivative test

At a critical point ($f'(x_0) = 0$), read the curvature:

curvature	local shape	classification
$f''(x_0) > 0$	smile — curve sits <i>above</i> its flat tangent	local min
$f''(x_0) < 0$	frown — curve sits <i>below</i> its flat tangent	local max
$f''(x_0) = 0$	test is silent	look further

Our worked example: $f''(x) = 6x$, so $f''(-1) = -6 < 0$ (max ✓) and $f''(1) = 6 > 0$ (min ✓) — matching the picture.

Honest footnote: $f(x) = x^4$ has $f'(0) = f''(0) = 0$, yet 0 is a perfectly good minimum. “Silent” means *silent*, not “no extremum”. (L9 upgrades this test to matrices — same fine print.)

Training halts where $\mathcal{L}'(\theta_0) = 0$ and $\mathcal{L}''(\theta_0) = 8$. What does the loss look like locally?

A. a frown — local maximum

B. a smile: $\mathcal{L}(\theta) \approx \mathcal{L}(\theta_0) + 4(\theta - \theta_0)^2$

C. a smile: $\mathcal{L}(\theta) \approx \mathcal{L}(\theta_0) + 8(\theta - \theta_0)^2$

D. flat in every direction — cannot say more

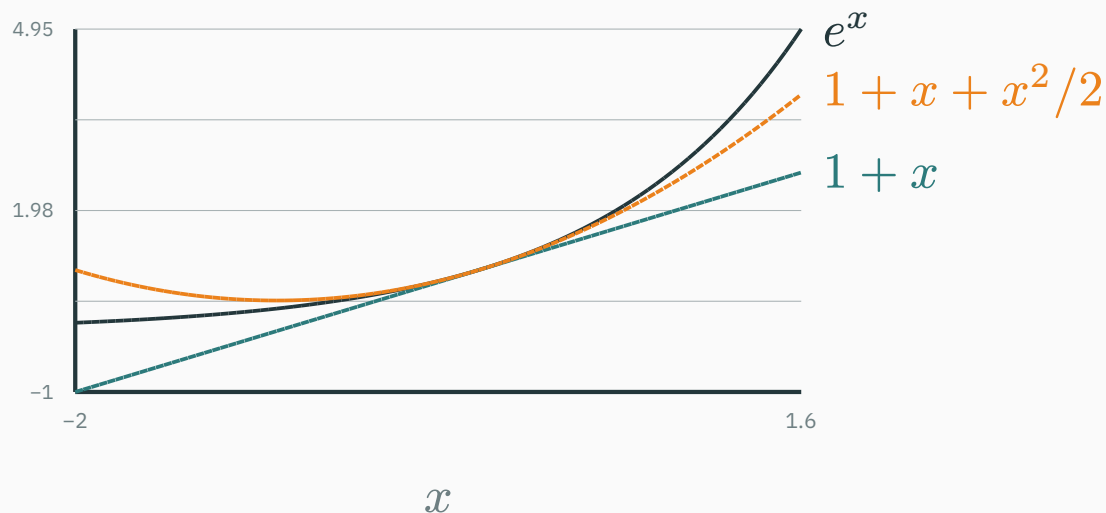
Correct: B — local min, $\mathcal{L}(\theta_0) + 4(\theta - \theta_0)^2$

Why: $\mathcal{L}'' > 0 \rightarrow$ smile $\rightarrow$ local minimum. The parabola's coefficient is $\frac{1}{2}\mathcal{L}''(\theta_0) = 4$, not 8 — that factor $\frac{1}{2}$ is Taylor's, and you are about to see exactly where it comes from.

Taylor: upgrading the line

The line is order one — keep going

The tangent matches f 's **value** and **slope** at x_0 . Next upgrade: also match the **curvature**. $f = e^x$ at 0:



The parabola hugs the curve visibly longer than the line. At $x = 1$: truth 2.718, line 2, parabola 2.5 — each order **halves the miss and more**.

Matching n derivatives

Order- n Taylor polynomial: the polynomial that agrees with f at x_0 in value and the first n derivatives — $n + 1$ conditions, $n + 1$ coefficients, one winner.

Why the coefficient of $(x - x_0)^k$ is $f^{(k)}(x_0)/k!$: differentiate $(x - x_0)^k$ exactly k times and the exponents come down as factors:

$$\frac{d^k}{dx^k} (x - x_0)^k = k \cdot (k - 1) \cdot \dots \cdot 1 = k! \quad \Rightarrow \quad \text{coefficient} = \frac{f^{(k)}(x_0)}{k!}$$

The $k!$ in the denominator is a **repair factor**: it cancels exactly this $k!$, so the k -th derivatives match.

Your checkpoint's $\frac{1}{2}$ was this factor at $k = 2$: coefficient = $f''(x_0)/2! = f''/2$.

The Taylor formula

$$f(x_0 + h) \approx f(x_0) + f'(x_0) h + \frac{f''(x_0)}{2!} h^2 + \frac{f'''(x_0)}{3!} h^3 + \dots + \frac{f^{(n)}(x_0)}{n!} h^n$$

- Order 0: a constant (stay where you are) · order 1: the local line · order 2: the local parabola...
- Each term is a **correction** the previous order couldn't see.

Near x_0 , Taylor's formula approximates a smooth function by a polynomial whose coefficients are its derivatives at x_0 .

★ Second-order Taylor of $\cos x$, step 1: harvest

D DERIVATION

Everything Taylor needs about $\cos$ at $x_0 = 0$ is three numbers. Harvest them:

$$f(x) = \cos x \quad \rightarrow \quad f(0) = 1$$

$$f'(x) = -\sin x \quad \rightarrow \quad f'(0) = 0$$

$$f''(x) = -\cos x \quad \rightarrow \quad f''(0) = -1$$

Keep harvesting and a pattern locks in: $f'''(0) = \sin 0 = 0$, $f^{(4)}(0) = \cos 0 = 1$, then the cycle $1, 0, -1, 0$ repeats forever. Every *odd* derivative dies at 0 — remember this.

★ Step 2: assemble, then test on numbers

Drop the three harvested numbers into the formula:

$$\cos x \approx 1 + 0 \cdot x + \frac{-1}{2} x^2 = 1 - \frac{x^2}{2}$$

Test at $x = 0.5$, where we can check everything by hand:

$$1 - \frac{0.25}{2} = 0.875 \quad \text{vs} \quad \cos(0.5) = 0.87758... \quad \rightarrow \quad \text{error} = 0.00258$$

Compare with the first term we dropped: the **next** Taylor term is $x^4/24$, and at $x = 0.5$ it equals $(0.5)^4/24 = 0.00260$.

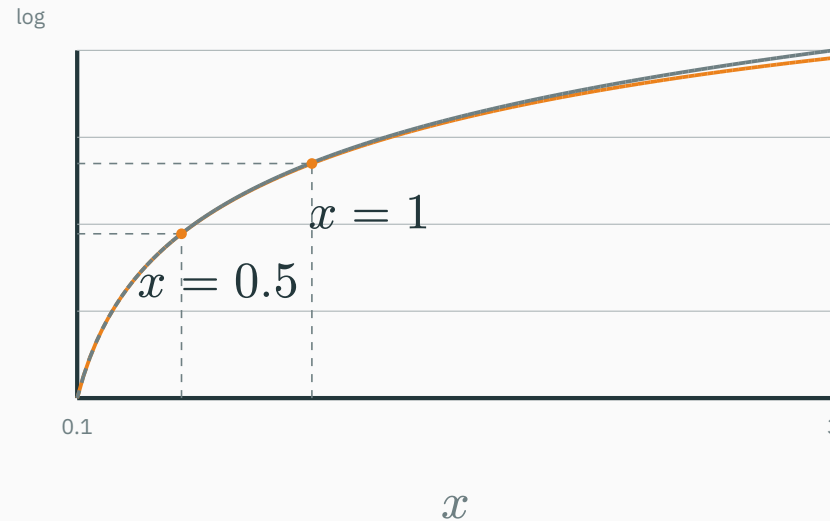
Here the error is almost the first omitted term, $x^4/24$. This numerical check is consistent with the Taylor remainder bound.

★ Step 3: the error, visualized

D DERIVATION

V VISUAL

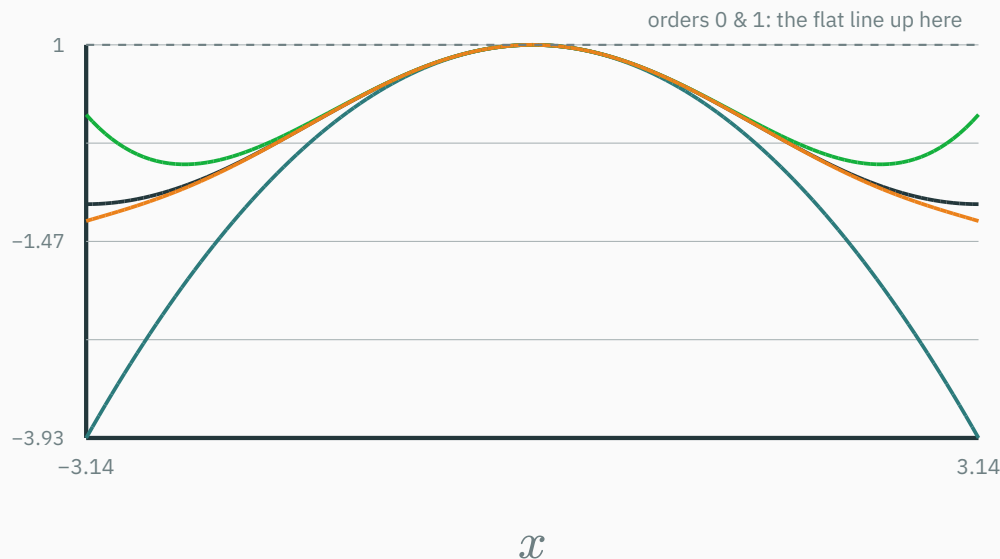
$|\cos x - (1 - x^2/2)|$ against the dropped term $x^4/24$ — log scale, six decades:



error of $1 - x^2/2$ · dashed: $x^4/24$ — glued together until $x \approx 1.5$

Quartic error: halve the distance $x \rightarrow$ the error falls $2^4 = 16 \times$. Second-order replacements are *absurdly* accurate close in.

The full build-up: $\cos x$, order by order



$\cos x$ · orders 0 & 1 · order 2 · order 4 · order 6

Orders 0 and 1 **coincide** — the slope at 0 is zero, so the best line is flat (every odd derivative died in step 1).

Each even order hugs the curve further out: the replacement's **trust region grows** with the order.

The build-up in code

```
from math import cos, factorial
def T(x, n):                                # order-n Taylor of cos at 0
    return sum((-1)**k * x**(2*k) / factorial(2*k)
               for k in range(n // 2 + 1))
for n in [0, 2, 4, 6]:
    print(n, T(1.0, n), f"{abs(cos(1.0) - T(1.0, n)):.4g}")
# 0  1.0                0.4597
# 2  0.5                0.0403
# 4  0.5416666666666666  0.001364
# 6  0.5402777777777777  2.453e-05
```

At $x = 1$, each added pair of terms cuts the error by a growing factor: $\times 11$, $\times 30$, $\times 56$. Six terms already give $\cos(1)$ to 4 digits. Real libraries combine polynomial approximations with **range reduction** rather than summing this Taylor series at arbitrary inputs.

The order-2 error of $\cos$ at $x = 0.2$ is about 6.7×10^{-5} . Doubling to $x = 0.4$ makes the error roughly...

A. $2 \times$ bigger

B. $4 \times$ bigger

C. $8 \times$ bigger

D. $16 \times$ bigger

Correct: D — $16 \times$ — **quartic scaling**

Why: Error $\approx x^4/24$, so doubling x multiplies it by $2^4 = 16$. (Check: at 0.4 the true error is $1.06 \times 10^{-3} \approx 15.9 \times$ bigger.) General rule: drop terms above order n , and the error scales like h^{n+1} .

Where the local approximation breaks

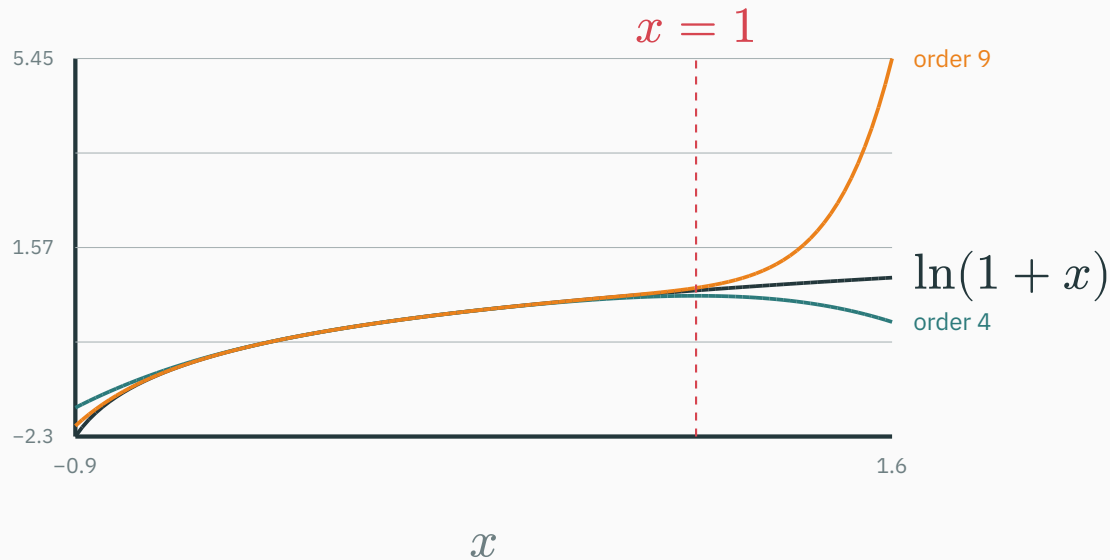
$\ln(1+x)$: a series with a boundary

Same game, new function: $\ln(1+x)$ at 0 has the series $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ Watch its partial sums at two inputs:

	S_1	S_2	S_3	S_4	S_5	S_6	truth
$x = 0.5$	0.5	0.375	0.417	0.401	0.407	0.405	$\ln 1.5 = 0.405$ ✓
$x = 2$	2	0	2.67	-1.33	5.07	-5.6	$\ln 3 = 1.099$ never

At $x = 0.5$ the sums settle beautifully. At $x = 2$ they **oscillate harder with every term** — the series diverges, even though $\ln 3$ itself is perfectly finite.

Watch it break



Inside $|x| < 1$: higher order = better, everywhere. Past $x = 1$ the orders **disagree violently** — order 9 rockets up, order 4 dives — while $\ln(1+x)$ itself barely changes.

The radius of convergence

Why 1? The series was built entirely from data **at $x = 0$** — and $\ln(1 + x)$ blows up at $x = -1$ (it plunges to $-\infty$).

For $\ln(1 + x)$ about 0, the singularity at $x = -1$ gives radius 1: the series converges for $|x| < 1$ and diverges for $|x| > 1$.

The failure need not occur where the real function itself is undefined: the singularity at -1 limits this expansion in both real directions, so it also fails at $+2$.

General fact (stated, not proved): for an analytic function, the radius is set by the nearest singularity in the **complex** plane. Smoothness alone does not guarantee that a Taylor series equals the function; the optional final slide shows why.

And $\cos$ never breaks

$\cos$ has no singularity anywhere — its radius is ∞ . The mechanism: **factorials beat powers**. Terms $x^k/k!$ at the distant point $x = 10$:

k	10	20	30	40
$10^k/k!$	2756	41	0.0038	1.2×10^{-8}

The terms **rise first** — then $k!$ catches up and crushes them. Enough terms deliver $\cos(10)$ to any accuracy you like.

Both behaviours occur: e^x , $\sin$, $\cos$ converge everywhere; $\ln(1+x)$ and $\frac{1}{1-x}$ have a finite radius. Always ask *where* a local replacement is valid — in ML we stay close to x_0 , which is why Taylor serves us so well.

What this buys ML

Apply the quadratic model to the opening loss

Zoom near a minimum θ^* and Taylor-expand the loss there:

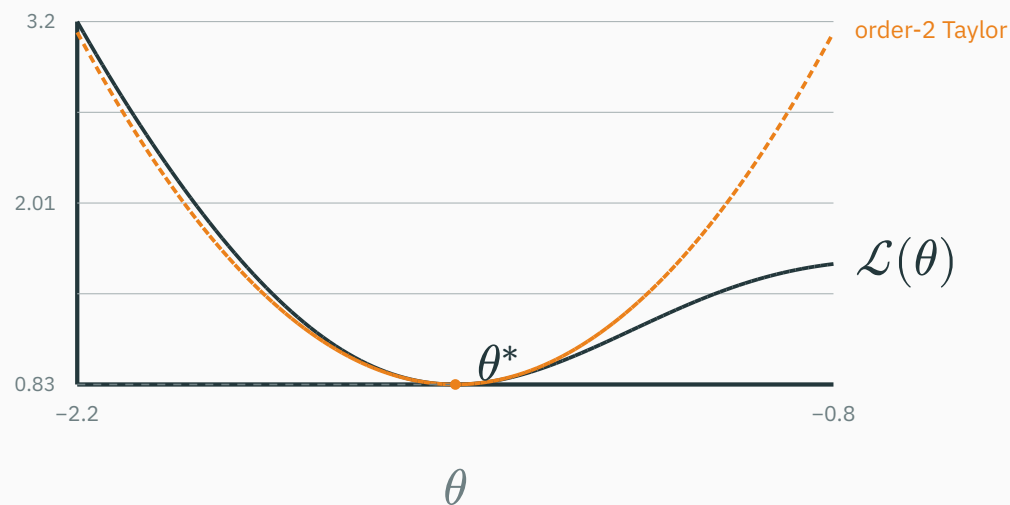
$$\mathcal{L}(\theta) \approx \mathcal{L}(\theta^*) + \underbrace{\mathcal{L}'(\theta^*)}_{=0 \text{ at a minimum}} (\theta - \theta^*) + \frac{1}{2} \mathcal{L}''(\theta^*) (\theta - \theta^*)^2$$

At a minimum the **line term is dead**. If $\mathcal{L}''(\theta^*) > 0$, the first surviving correction is quadratic, so the loss locally looks like a parabola. If the second derivative also vanishes, higher orders decide.

Anywhere else, $\mathcal{L}' \neq 0$: the line term dominates all others for small steps — which is why zooming elsewhere gave a **line**.

Wiggly $\rightarrow$ parabola at minima, line elsewhere: not a coincidence — it's Taylor, order by order.

Near its minimum ($\theta^* = -1.497$, $\mathcal{L}^* = 0.828$, $\mathcal{L}''(\theta^*) = 9.39$), compare the opening loss with the quadratic model $\mathcal{L}^* + 4.69 (\theta - \theta^*)^2$:



Near θ^* the quadratic is an accurate local model of this loss. Later optimizers use either this local quadratic or its linear part.

Where L7 is used later

The same local approximations reappear in later lectures:

lecture	what it does with today's Taylor
L8–L9	slope $\rightarrow$ gradient, curvature $\rightarrow$ Hessian: same formulas, more dimensions
L16	convexity: every local minimum is global; the Hessian is PSD where defined
L17	gradient descent derived : trust the line, step downhill
L18	Newton's method derived : trust the parabola, jump to its bottom

When L17 writes $\theta_{t+1} = \theta_t - \eta \mathcal{L}'(\theta_t)$, it is quoting today's local line; when L18 divides by $\mathcal{L}''$, it is quoting today's parabola.

Summary & what's next

Lecture 7 — summary

- **Derivative = sensitivity**: the ratio $\Delta f/h$ stabilizes — 7, 6.1, 6.01, ... $\rightarrow 6$.
- **Local line**: $f(x_0 + h) \approx f(x_0) + f'(x_0)h$ — the only error that beats h .
- **Chain rule**: pipelines multiply local ratios ($3 \times 14 = 42$); ML is a deep pipeline.
- **Flat + curvature**: $f' = 0$ finds candidates; the sign of f'' classifies them — usually.
- **Taylor**: match n derivatives, repair by $k!$; ★ $\cos x \approx 1 - x^2/2$, error $\approx x^4/24$.
- **Radius**: for analytic functions, the Taylor series has a convergence radius; $\ln(1 + x)$ about 0 has radius 1, while $\cos$ converges everywhere.
- **At a smooth minimum**: the linear term vanishes and the Hessian determines the local quadratic shape used in L16–L18.

Watch: 3b1b *Essence of Calculus* ch. 1–4 + the Taylor video. **Read before L8:** MML §5.1. **T4** (after L8): derivative and chain-rule drills in NumPy.

Practice problems

Try on paper; verify against the T4 notebook.

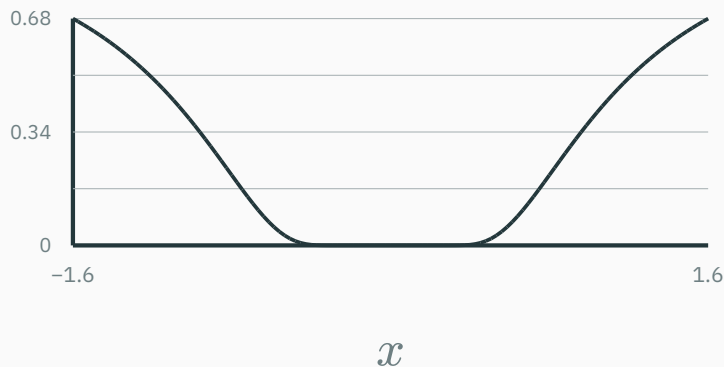
1. Build the nudge table for $f(x) = 1/x$ at $x_0 = 2$ with $h = 0.1, 0.01, 0.001$. What is the ratio heading toward?
2. Estimate $e^{0.1}$ with the local line at 0, then with the order-2 Taylor. Compare both to 1.10517...
3. For $y = (2x - 1)^3$ at $x = 1$: tabulate $\Delta x \rightarrow \Delta u \rightarrow \Delta y$ for $\Delta x = 0.01$ and verify the local ratios multiply to $dy/dx = 6$.
4. Find and classify every critical point of $f(x) = x^3 - 6x^2 + 9x + 1$.
5. Derive the order-3 Taylor of $\sin x$ at 0, evaluate it at $x = 0.3$, and bound the error by the first dropped term. Why do only odd powers appear?
6. The $\ln(1 + x)$ series diverges at $x = 2$ although $\ln 3$ is finite. In one sentence, what sets the radius, and where is the offending point?



The function Taylor cannot see

★ OPTIONAL

$f(x) = e^{-1/x^2}$ (with $f(0) = 0$) is **smooth everywhere** — yet at 0, every derivative equals 0:



Its Taylor series at 0 is $0 + 0x + 0x^2 + \dots \equiv 0$ — it converges beautifully, **to the wrong function**. The curve is flatter at 0 than any polynomial; Taylor is blind to it.

Moral: *smooth* $\neq$ *analytic* (faithfully told by its Taylor series). Our losses never hit the distinction — but it is why the radius slide said *analytic*, not just *smooth*.

Calculus is the art of replacing a function with a line.

Next: **Gradients & the Geometry of Surfaces** — the same replacement idea, in more dimensions: the line becomes a plane, the slope becomes a vector.